CHAPTER 01

Classifying Life's Diversity

There are over 8 million kinds of living things on Earth — from giant whales to germs too small to see. How do scientists make sense of them all? By sorting life into groups, just like sorting a giant playlist. Let's learn how.

📗 **Georgia Standard S7L1**

WHAT YOU'LL LEARN

1.1 What Makes Something "Alive"?

> Before we can sort living things into groups, we need to answer a sneaky question: what counts as **living** in the first place? A rock just sits there. A dog runs, eats, and grows. But what about a seed, or a virus? Scientists use a checklist of traits that **every** living thing shares.

An object is considered **alive** only if it does **all** of the things on this list. Miss even one, and it's not a living organism. A car uses energy and "responds" when you turn the wheel — but it can't grow or reproduce, so it's not alive.

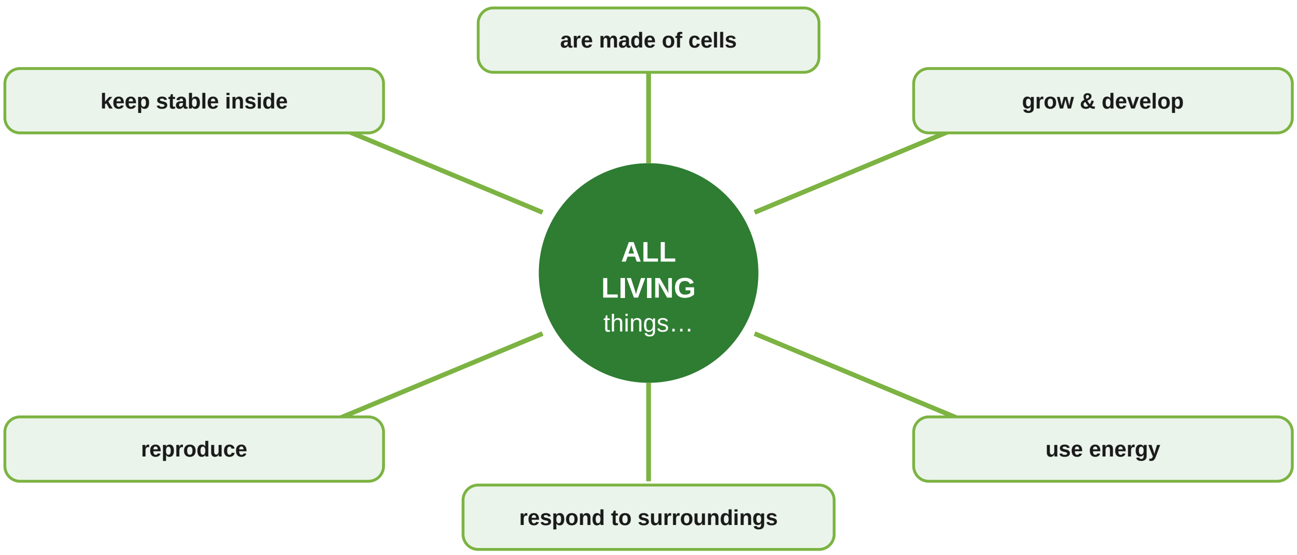

Figure 1.1 — The six characteristics shared by all living things.

! Did You Know?

The largest living thing on Earth is a fungus in Oregon. It spreads underground across 3.5 square miles — bigger than 1,600 football fields — and may be over 2,400 years old!

✗ Common Mix-Up

Movement ≠ Life. A river moves and a flame "grows," but neither is alive. And many living things (like trees and coral) barely move at all. Use the full checklist, not just one trait.

1.2 Why Do Scientists Classify Life?

> Imagine a grocery store where everything was dumped into one giant pile — no aisles, no signs. Finding milk would take hours. **Classification** is the science of sorting living things into organized groups so we can study, name, and understand them.

Scientists who classify organisms are called **taxonomists**, and the science itself is called **taxonomy**. Classifying life helps us in three big ways:

- **It organizes** millions of species into a system everyone can use.
- **It reveals relationships** — organisms in the same group share an ancestor.
- **It helps prediction** — if a new animal has fur and feeds milk to its babies, we instantly know a lot about it before studying it closely.

🔑 **Key Idea**

Organisms are grouped by **shared characteristics** — the traits they have in common. The more traits two organisms share, the more closely related they usually are.

1.3 A History of Classification

> People have been sorting living things for thousands of years, but the system has changed a lot as we learned more — especially after the invention of the microscope.

When	Who	What They Did
~350 BCE	Aristotle	Sorted life into just two groups: **plants** and **animals**. Animals were grouped by where they lived — land, water, or air.
1735	Carolus Linnaeus	Created the system we still use today. He grouped organisms by **physical characteristics** and gave each a two-word scientific name.
1600s–now	Microscope users	Discovered tiny organisms (bacteria, protists) that didn't fit "plant" or "animal." New kingdoms had to be added.
Today	Modern scientists	Use **DNA** and cell structure — not just looks — to group organisms into the **six-kingdom system**.

! Why the System Keeps Changing

Early scientists could only group organisms by what they could **see**. Two animals that looked alike were placed together — even if they weren't really related. Today, DNA lets us look **inside** the cell, so classification is far more accurate. Science improves as our tools improve.

1.4 The Levels of Classification

Linnaeus's system sorts every organism through **eight levels**, from the broadest group (Domain) down to the most specific (Species). As you move down, the groups get smaller and the organisms inside become more and more alike.

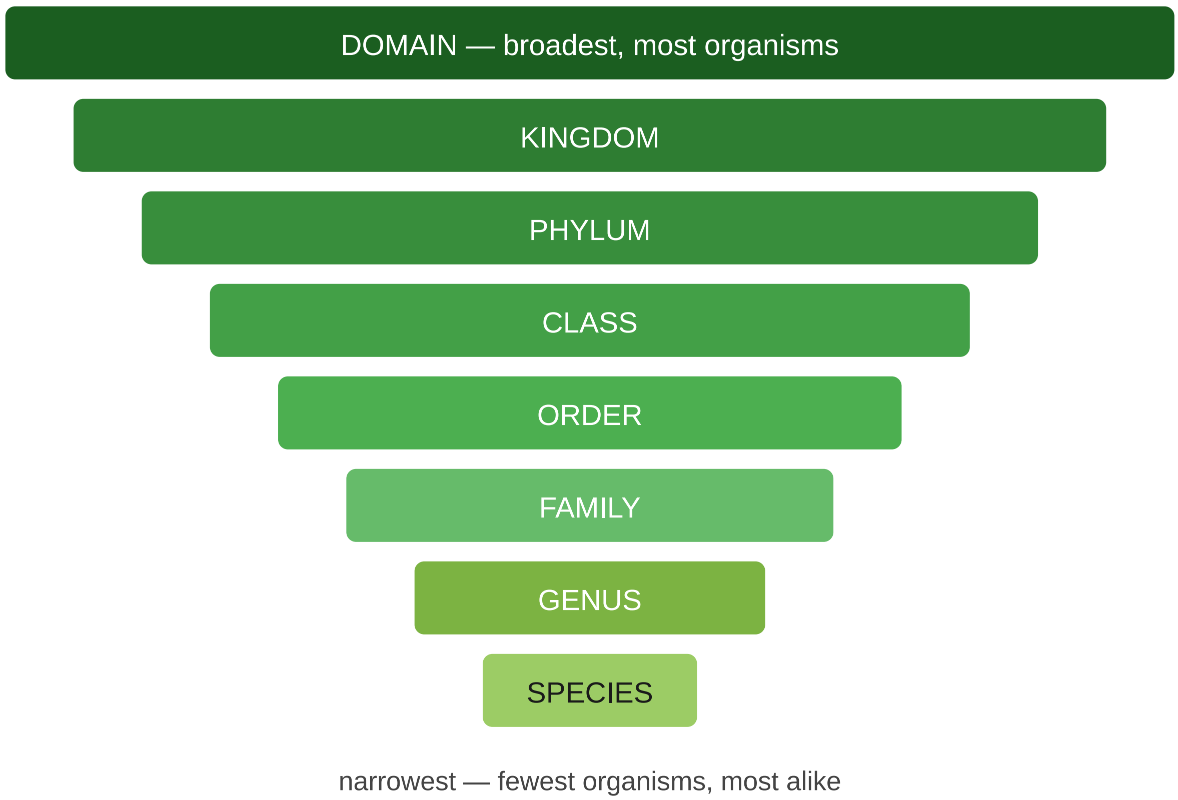

Figure 1.2 — The levels of classification, broadest to narrowest.

⭐ **Memory Trick**

Use this silly sentence to remember the order — the first letter of each word matches each level:
Dear **K**ing **P**hilip **C**ame **O**ver **F**or **G**ood **S**oup
= **D**omain, **K**ingdom, **P**hylum, **C**lass, **O**rder, **F**amily, **G**enus, **S**pecies.

Following One Animal Down the Levels

Here's how a single gray wolf gets classified all the way down. Notice how each step narrows the group:

Level	Gray Wolf belongs to…	Who else is in this group?
Kingdom	Animalia	All animals — insects, fish, humans
Phylum	Chordata	Animals with a backbone
Class	Mammalia	Mammals — dogs, cats, whales, humans
Order	Carnivora	Meat-eaters — bears, lions, seals
Family	Canidae	Dog family — foxes, coyotes, jackals
Genus	*Canis*	Wolves, dogs, coyotes
Species	*Canis lupus*	Only gray wolves

1.5 The Six Kingdoms of Life

> At the top of the system, all living things are sorted into **six kingdoms**. Scientists decide which kingdom an organism belongs to by asking three questions: Is it made of one cell or many? Do its cells have a nucleus? And how does it get food?

🦠 **Archaea**	🦠 **Bacteria**	🟢 **Protists**
1 cell, no nucleus makes own/absorbs *hot-spring microbes*	1 cell, no nucleus absorbs food *strep, E. coli*	1 or many cells varies *amoeba, algae*
🍄 **Fungi**	🌿 **Plants**	🐾 **Animals**
many cells absorbs food *mushroom, mold*	many cells makes own food *trees, moss*	many cells eats food *dog, human, ant*

⭐ The Big Split

The first divide is whether a cell has a **nucleus**. Archaea and Bacteria are **prokaryotes** (no nucleus). The other four kingdoms are **eukaryotes** — their cells keep DNA inside a nucleus.

! Did You Know?

Archaea were only recognized as their own kingdom in the 1970s. Many live in extreme places — boiling vents, super-salty lakes — where almost nothing else survives.

How to Tell the Kingdoms Apart

Kingdom	Cell Type	# of Cells	Gets Food By…
Archaea	Prokaryote (no nucleus)	One	Making or absorbing
Bacteria	Prokaryote (no nucleus)	One	Absorbing
Protists	Eukaryote (nucleus)	One or many	Varies — many ways
Fungi	Eukaryote (nucleus)	Mostly many	Absorbing (decomposers)
Plants	Eukaryote (nucleus)	Many	Making food (photosynthesis)
Animals	Eukaryote (nucleus)	Many	Eating other organisms

1.6 Tools for Sorting Life

> When scientists (or students!) need to identify an unknown organism, they use a **dichotomous key** — a step-by-step list of either/or questions. Each answer sends you down a different path until you reach the organism's name.

The word *dichotomous* means "divided into two." At every step you pick between two choices, splitting the possibilities in half until only one organism is left.

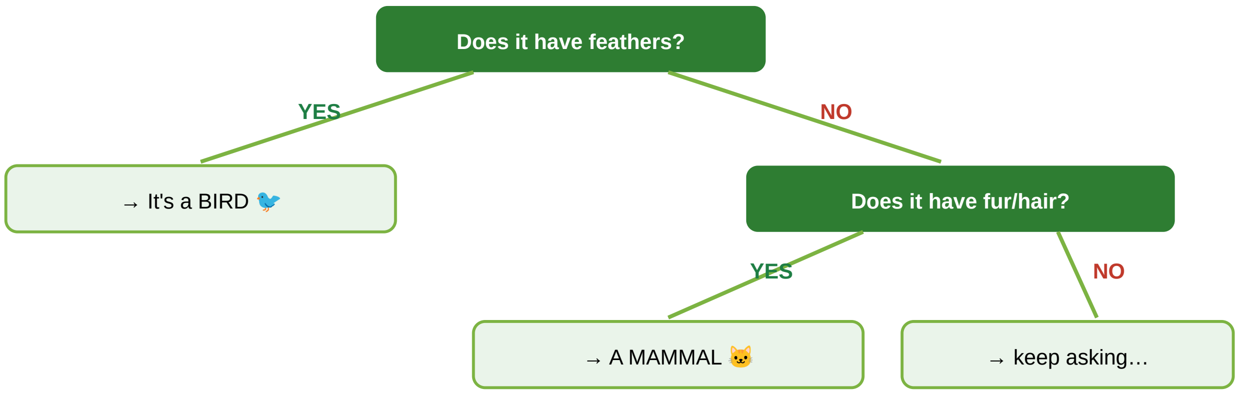

Figure 1.3 — A simple dichotomous key. Each question has exactly two answers.

⭐ How to Use a Key

- Always start at **step 1**.
- Read both choices before deciding.
- Pick the choice that matches, then follow it to the next step.
- Keep going until you reach a name — not another question.

1.7 Naming Living Things

> A robin might be called a "robin" in English, but it has different names in every language. To avoid confusion, scientists everywhere use one shared naming system called **binomial nomenclature** — meaning "two-word naming."

Every species gets a two-part scientific name made from its **Genus** and **species**. This name is always written in a special way:

📝 The Rules of Scientific Names

- The **Genus** is written first and is **Capitalized**.
- The **species** is written second and is lowercase.
- The whole name is *italicized* (or underlined when handwritten).

Example: humans are *Homo sapiens*. House cats are *Felis catus*.

! Did You Know?

Scientific names are often Latin or Greek and describe the organism. *Tyrannosaurus rex* means "tyrant lizard king" — a pretty good name for a giant predator!

✗ Common Mix-Up

Don't capitalize the species name. It's *Homo sapiens*, never *Homo Sapiens*. Only the genus gets a capital letter.

1.8 Putting It All Together

You've now seen how scientists make sense of life's huge variety. They start by checking whether something is **alive**, then sort living things into a system that goes from broad **domains and kingdoms** down to a single **species**. They use **dichotomous keys** to identify organisms and **scientific names** so everyone, everywhere, means the same thing.

Big Idea	What to Remember
Characteristics of life	All living things share 6 traits — cells, energy use, growth, reproduction, response, and staying stable inside.
Classification	Sorting organisms by shared traits; the science is taxonomy.
History	Aristotle → Linnaeus → microscopes → DNA. The system improved as tools improved.
Levels	Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species.
Six kingdoms	Archaea, Bacteria, Protists, Fungi, Plants, Animals.
Naming	Two-word scientific names: *Genus species*.

⭐ **Standard Check — S7L1**

You can now **(a)** build a model that groups organisms by common characteristics, and **(b)** explain how classification grew from simple physical-trait systems into today's six-kingdom system. That's the whole standard — nice work!